

Three Notions of Linearization for a Discrete Ising System: Operator, Statistical, and Projection-Based Approximations

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Abstract—For stochastic, high-dimensional nonlinear systems, the notion of “linearization” is ambiguous and depends on how state, dynamics, and observables are represented. In this project, we study a discrete-time Ising model with Glauber dynamics and ask how three distinct linear representations differ in their fixed points, stability properties, and retained structure. We construct each representation from the same underlying dynamics and analyze how aggregation and coarse-graining alter equilibria and bifurcation behavior. Our analysis showcases the idea that linearization is not a single algorithm (a first-order approximation at a point) but a modeling choice that trades off between exactness, complexity reduction, and interpretability, with concrete consequences for captured behavior.

I. MOTIVATION

Nonlinear systems theory traditionally emphasizes local linearization via the Jacobian. However, for stochastic and high-dimensional systems, especially those arising in statistical physics and multi-agent decision processes, multiple nonequivalent notions of “linearization” coexist. These include exact linear representations over probability simplices, heuristic approximations of variables, and data-driven reduced-order linear models obtained by projection.

The Ising model [1] provides a canonical nonlinear, stochastic, and discrete-time system in which these notions can be compared in a controlled setting. This project aims to formalize and contrast three linearization paradigms applied to a discrete-time Ising dynamical system, highlighting what structural information is preserved or lost by each approximation.

II. INTRODUCTION AND BACKGROUND

The Ising model, commonly used in materials physics, consists of an undirected graph $G = \{V, E\}$, where a set of vertices or nodes V represent atoms and edges $E = \{(i, j) : i, j \in V\}$ represent interactions between atoms. Typically, the graph forms a regular lattice (meaning there exists a translational symmetry in space), as the behavior of a specific crystal or alloy with a regular structure is studied. Each atom has a spin, which determines its magnetic alignment, and this spin is either aligned (+1) or anti-aligned (-1) with any reference spins (including its neighbors). Atoms have an associated energy level, which is higher for the spin that conflicts with their alignment preferences (depending on the alignment of their neighbors and any external forces), and

they generally prefer their spins to be in the minimum-energy alignment. Although in real materials, atoms may simultaneously flip spins, and dynamics occur through various quantum and other interactions, the Ising model predicts very well the overall behavior of the material even when update computations are performed one atom at a time in a randomized order.

One property of interest of the material is coupling, or the nature and strength of the edge weights J_{ij} or interactions between atoms. When the coupling is positive, spins prefer alignment; these are ferromagnetic materials. Many materials are non-ferromagnetic, with $J \approx 0$, and a few are even anti-ferromagnetic, with $J < 0$. The material is also exposed to an external magnetic field with alignment h , which encourages all atoms to favor a particular alignment, dependent on other conditions. Finally, the material has a temperature T , which is essentially a measure of how willing atoms are to have an alignment with a higher energy state (zero temperature corresponds with no free energy to assume the higher energy of the two possible spins).

The Ising model displays several key properties that make it an interesting and useful model for materials science. First, it captures the observation that regions or *domains* of atoms tend to align at low energies, and when these regions are biased toward a certain alignment, the overall material exhibits a mean magnetization. As energy increases, these patterns dissolve into a chaotic noise of uncorrelated spins, even between neighboring atoms in the limit of high temperature. Most importantly, between these two modes, at the phase transition, there exists a transitional or critical state where correlation between spins is still influential, but chaos has begun to infiltrate the system, such that perturbations to one part of the system have an influence not only in the atom’s local domain, but can propagate across the entire lattice, essentially destroying the compartmentalization of the domains but retaining some concept of locality for each individual atom. The resulting structure has a distinctive form, with constantly evolving, merging, splitting, and even translational movement of domains of every size, and the sizes of these domains obey a power-law-like distribution. The main consequence of this structure is its scale-free nature; structure exists at every spatial scale, and the domains form a fractal such that any sub-region of the process is essentially similar to the entire material [2]. The paradox at criticality is that although there is distinct structure and observable domains, these domains are of wildly differing sizes, almost as though the material has coordinated large-scale dynamics with small-scale dynamics.

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The Ising model is famous for its universality. Universality is the property of a model to predict the behavior for a wide range of systems. Although the Ising model was first proposed to model the spins of atoms in a material, the properties of the model, including phase transitions and a critical [3], it and essentially isomorphic models are used to model percolation processes [4], phase transitions [5], and forest fires [6], to name only a few. All of these modifications and settings display the same scale-free behavior, phase transitions, and criticality.

The Ising model is also often a first concrete introduction to the concept of *emergence*. Emergence is the quality of a system composed of components that interact according to a set of rules, and whose composition exhibits behavior that the individual components cannot exhibit [7]. In other words, each component does not work as a good approximate model of the entire system. These systems often exist at a critical point between too few rules and too many rules, such as Conway’s Game of Life [8] and chess on an infinite board, which are both Turing-complete. The author’s opinion is that this property of criticality (which often does manifest as Turing-completeness) is essential to sufficiently interesting systems and plays a complementary role in the philosophy of modeling such systems by intelligent agents (see Appendix).

Discussion on emergence from structured composition is fundamentally a discussion about scale. The phenomenon of emergence is often accompanied by a Markov blanket of scale; in fact, the entire philosophy of reductionism stems from the assumption that large and small scales are separated by a Markov blanket of only a few key features. For example, the properties of a gas made up of moles of atoms are described best not by the quantum field theory governing individual atoms, but by the volume the atoms fill, the number of atoms, the mean kinetic energy of the atoms (temperature), and the frequency of collisions, including with the boundaries of the volume (pressure). These quantities are related by an ideal gas law, which is much more tractable and descriptive of the gas as a whole than describing the states of each individual atom. This process of aggregating small-scale features into higher-scale laws is formalized in *renormalization* [9], [10], [11]. This approach goes by many names, including projection, course-graining, renormalization, emergence, abstraction, aggregation, mean-field theory, sufficient statistics, balanced truncation model reduction [12], Koopman observables [13], [14], [15], or simply “stepping back,” “zooming out,” or “seeing the big picture.”

On the other end of the spectrum of computability is the concept of linearity. Linearity is the property that some algebraic structures have, which allows them to be decomposed into components, such that analysis can be performed on those (often simpler) components individually, and those analyses can be aggregated via an addition operation satisfying certain basic axioms [16]. This capability of breaking problems into smaller components, and that *any* combination of those components can be handled by simply knowing what to do with each component, makes these structures

very powerful. The power of linear representations is well illustrated by linear systems theory, where optimal controllers and even system identification are often performed efficiently using linear methods [17], as well as by the theory of linear PDEs, where existence, uniqueness, and stability can often be established using functional-analytic tools [18]. A slight relaxation of this property is convexity, where instead of any combination of these components, we have some set of boundary elements, and the space of all elements enclosed by those boundary elements is admissible under the problem at hand. This allows for efficient algorithms to search for solutions so long as we can characterize the boundary elements [19]. Our goal is then to reformulate the nonlinear and difficult problem of Ising dynamics into a convex (or ideally linear) learning and modeling problem.

III. SYSTEM DESCRIPTION

We consider an N -spin Ising system with configuration $s \in \{-1, +1\}^N$ evolving in discrete time according to Glauber dynamics [20] under spin-coupling strengths $J_{ij} \in [-1, +1]$, an external field $h \in \mathbb{R}$, and a temperature $T \geq 0$:

$$\mathbb{P}(s_i^{k+1} = +1 \mid s^k) = \sigma \left(\sum_j J_{ij} s_j^k + \mathcal{N}(h, T^2) \right),$$

where $\sigma(\cdot)$ is the logistic function and $\mathcal{N}(\mu, \sigma^2)$ is a normal distribution. We will always use the superscript k to denote time, and subscripts denote vector indices. This defines a discrete nonlinear stochastic map on the configuration space of the spins. In many scenarios, J will contain structure encoding the topology of the spins, such as atoms that are only influenced by their neighbors, as well as the properties of the material (some materials prefer anti-alignment, etc.).

IV. METHODOLOGY

The project investigates three representations of this system with the intention of producing a descriptive model of the dynamics, which can be described as linear. Most of these models are linear in the probabilities of the states, as the state dynamics themselves are nonlinear.

A. Exact MC / Operator Linearization

A Markov Chain (MC) is a probabilistic state evolution process where each of a collection of states has some probability of transitioning to another state during some time evolution [21], a discrete companion of Markov processes [22]. Markov decision processes are similar, but with an optimization process or decision-making agent in-the-loop. They are, by definition, *memoryless*, meaning there exists a temporal Markov blanket, or a set of sufficient statistics such that knowing these states provides all necessary information to continue state evolution analysis (i.e. if we know the present state, we can throw out all past states). These processes are said to be *time invariant* or stationary if the dynamics or state transition probabilities are the same for all time, *causal* if state evolution only depends on states in the past, never the future. An important feature of Markov

models is that they are linear in the probabilities of each state or configuration, and are representable by a stochastic state transition matrix [23] of conditional probabilities of the form $P_{ij} \triangleq \mathbb{P}(s_i^{k+1}|s_j^k)$, with state s and time k .

The system is represented as a Markov Chain over the probability simplex $\{p_s \in \mathbb{R}_{\geq 0}^{\{-1,+1\}^N} : \sum_{s \in \{-1,+1\}^N} p_s = 1\}$:

$$p^{k+1} = P(J, h, T)p^k,$$

where p^k is the distribution over each configurations s at time k and $P(J, h, T)$ is a column-stochastic transition matrix. This representation is fully linear but exponentially large. Fixed points correspond to stationary probability distributions.

This model retains the intricate patterns of the Ising model, such as the "islands within islands" power-law-like structures at criticality, at the expense of a state space exponential in the number of spins.

B. First-Principles Linear / Statistical Model

In this model, we represent the state of our MDP as a probability distribution over each spin being ± 1 . Each of the N spins has two associated nonnegative values, $p_i(+1) + p_i(-1) = 1$, stacked in a vector $p \in \mathbb{R}_{[0,1]}^{2N}$. This naturally partitions the following transition matrices into 2×2 blocks, each block corresponding to one atom.

We restrict or normalize the coupling, external field, and temperature to the ranges $J_{ij} \in [-1, +1]$, $h \in [-1, +1]$, $T \in [0, 1]$, respectively. The transition matrix $A(J, h, T)$ is:

$$\frac{1}{2}T\mathbf{1} + (1-T)\left(|h|\mathbf{H}(h) + (1-|h|)\mathbf{J}_*(J)\right)$$

Where $\mathbf{1}$ is the block-diagonal matrix with 2×2 blocks of ones on the diagonal blocks, and zeros on the off-diagonal blocks, which effectively homogenizes the probabilities of each spin to $p_i(+1) = p_i(-1) = 0.5$, $\forall i$. $\mathbf{H}(h)$ is also a block-diagonal matrix with the same structure, but the diagonals are:

$$\begin{bmatrix} \frac{1+h}{2} & \frac{1+h}{2} \\ \frac{1-h}{2} & \frac{1-h}{2} \end{bmatrix}$$

This adds a bias to each next-timestep probability that is independent of the previous probabilities (as the row vectors homogenize the previous distribution). Finally, $\mathbf{J}_*(J)$ is a block matrix of a similar shape, but the (i, j) th 2×2 block of $\mathbf{J}_*$ is:

$$\frac{1}{\sum_k |J_{ik}|_0} \begin{bmatrix} \frac{1+J_{ij}}{2} & \frac{1-J_{ij}}{2} \\ \frac{1-J_{ij}}{2} & \frac{1+J_{ij}}{2} \end{bmatrix}$$

with zero blocks where $J_{ij} = 0$. This operator is the only of the three where the rows are heterogeneous, and they have the effect of homogenizing the distribution when $J_{ij} = 0$, biasing toward alignment (pos. larger $\rightarrow$ pos. larger, etc.) when $J_{ij} > 0$, and biasing toward anti-alignment when $J_{ij} < 0$. The normalization constant ensures that the entire $\mathbf{J}_*$ matrix is row-stochastic by dividing out the number of nonzero blocks in that block-row, i.e. $|x|_0 \triangleq \mathbb{I}(x \neq 0)$,

the nonzero indicator or zero-norm. Row stochasticity is the required condition for each individual spin to always have cumulative probability one; column stochasticity is not required as it is each individual spin that requires a conservation of probability density, not the entire p vector.

The following heuristics motivate the dynamics of this simplified model:

- As $T \rightarrow 1$, the dynamics are drowned by the noise or temperature term and the expected spin is everywhere $\frac{1}{2}$.
- Else, as $h \rightarrow \pm 1$, the external field overwhelms the spins, and the expected spins tend toward ± 1 .
- Else, if the mean coupling is positive, spins will tend to be more like the average of their neighbors. Likewise, if the mean coupling is negative, spins will tend to be opposite the average of their neighbors.

This model has no extra parameters; however, J, h, T are in new units, which must be relearned from the data. I propose that the nonlinear transform determining these parameters from their counterparts in the full model might be somewhat helpful, but in a probabilistic model, homogenization of probabilities erases much of the interesting structure the Ising model captures, especially at the critical point, so it might not be entirely useful to learn the exact map for these new parameters.

C. Fokker-Planck / Projection Approximation

By projection, we mean projecting onto a lower-dimensional state space via coarse-graining. The hope is that, by tracking fewer features (and selecting sufficient statistics to be those features), we can make a working predictive model in those features without needing observability of all the low-level states at every time step [24].

Defining a flip rate

$$w_i(s) = \frac{1}{2} \left[1 - s_i \sigma(Js + \mathcal{N}(h, T^2)) \right]$$

And denoting the operator that flips the spin of atom i as $F_i s$, we obtain the master equation (a discrete equivalent of the Fokker-Planck equation) [25] for these dynamics:

$$\frac{\partial P(s, t)}{\partial t} = \sum_i [w_i(F_i s)P(F_i s, t) - w_i(s)P(s, t)]$$

To obtain a true Fokker-Planck equation, we can then aggregate or coarse-grain discrete microscopic states into continuous macroscopic observables such as magnetization [26]:

$$m^k = \frac{1}{N} \sum_i s_i^k.$$

We could have equivalently used a different sufficient statistic, such as the proportion of atoms with positive spin, which is equal to $(m+1)/2$. Further assuming that the material is homogeneous (i.e., that the coupling is the same throughout, so $\bar{J}$ is the mean coupling), we can define the associated Fokker-Planck equation as:

$$\frac{\partial P(m, t)}{\partial t} = -\frac{\partial}{\partial m} [A(m)P(m, t)] + \frac{1}{2} \frac{\partial^2}{\partial m^2} [B(m)P(m, t)]$$

where

$$A(m) = \sigma(\bar{J}m + h) - m$$

is the drift, and

$$B(m) = \frac{2}{N} \left[1 - m\sigma(\bar{J}m + h) \right]$$

is the diffusion.

Under standard assumptions, the dynamics admit a mean-field approximation

$$m^{k+1} \approx \sigma(Jm^k + h),$$

Fixed points and bifurcations are analyzed in this reduced space, especially near the critical point [27], [28].

Similar to most observed systems, we modeled the aggregation of many states s as a linear projection C , mapping the full probability state to a few important statistics (in this case, one):

$$\begin{aligned} s^{k+1} &= As^k + Bh \\ m^k &= Cs^k \end{aligned}$$

A linear model

$$m^{k+1} \approx \tilde{A}m^k + \tilde{B}h$$

can then be fit using simulation data or derived by approximating the Fokker-Planck or heuristic solutions above. The matrix C is interpreted as a renormalization or coarse-graining operator [28], [10]; it computes the sufficient statistics for building a predictive model.

V. ANALYSIS GOALS

The project will:

- Quantify approximation error introduced by aggregation
- Compute and compare fixed points across representations
- Study stability and bifurcation structure under varying J , h , and T .
- Interpret C as an observability or renormalization map

Python will be used to:

- Simulate Glauber dynamics
- Construct transition matrices for small N
- Estimate reduced linear models
- Generate figures illustrating fixed points and bifurcations

VI. EXPECTED CONTRIBUTIONS

This project will demonstrate that “linearization” is not a single operation but a family of approximations with distinct mathematical and physical interpretations. The comparison highlights trade-offs between exactness, dimensionality, and interpretability, providing a nonlinear-systems perspective on statistical physics models.

REFERENCES

- [1] E. Ising, “Beitrag zur theorie des ferromagnetismus,” *Zeitschrift für Physik*, vol. 31, no. 1, pp. 253–258, 1925.
- [2] D. Stauffer and A. Aharony, *Introduction to percolation theory*. Taylor & Francis, 2018.
- [3] M. Kardar, “Statistical mechanics i: Statistical mechanics of particles, fall 2007,” *Massachusetts Institute of Technology. Department of Physics*, 2007.
- [4] C. Alves, A. Hrušková, and V. Winstein, “Percolation: a mathematical phase transition.” <https://www.youtube.com/watch?v=a-767WnbaCQ#t=5m23s>.
- [5] V. Winstein and G. Sanderson, “Simulating and understanding phase change | guest video by vilas winstein.” <https://www.youtube.com/watch?v=itRV2jEtV8Q#t=12m05s>.
- [6] J. Moore, C. Mebius, and D. Muller, “You’ve (likely) been playing the game of life wrong.” https://www.youtube.com/watch?v=HBluLfX2F_k#t=18m55s.
- [7] Emergent-Garden, “Emergent complexity.” <https://www.youtube.com/watch?v=0HqUYpGQIfs>.
- [8] M. Games, “The fantastic combinations of john conway’s new solitaire game “life” by martin gardner,” *Scientific American*, vol. 223, no. 120-123, p. 11, 1970.
- [9] Broken-Symmetries, “Renormalization: Why bigger is simpler.” <https://www.youtube.com/watch?v=9vFbyHNz-8g#t=3m56s>.
- [10] N. Goldenfeld, *Lectures on phase transitions and the renormalization group*. CRC Press, 2018.
- [11] P. W. Anderson, “More is different: broken symmetry and the nature of the hierarchical structure of science,” *Science*, vol. 177, no. 4047, pp. 393–396, 1972.
- [12] S. Gugercin and A. C. Antoulas, “A survey of model reduction by balanced truncation and some new results,” *International Journal of Control*, vol. 77, no. 8, pp. 748–766, 2004.
- [13] C. W. Rowley, I. Mezić, S. Bagheri, P. Schlatter, and D. S. Henningson, “Spectral analysis of nonlinear flows,” *Journal of fluid mechanics*, vol. 641, pp. 115–127, 2009.
- [14] M. Budišić, R. Mohr, and I. Mezić, “Applied koopmanism,” *Chaos: An Interdisciplinary Journal of Nonlinear Science*, vol. 22, no. 4, 2012.
- [15] P. Walters, *Ergodic theory—introductory lectures*, vol. 458. Springer, 2007.
- [16] G. Strang, *Introduction to linear algebra*. SIAM, 2022.
- [17] D. G. Luenberger, *Optimization by vector space methods*. John Wiley & Sons, 1997.
- [18] L. C. Evans, *Partial differential equations*, vol. 19. American mathematical society, 2022.
- [19] S. Boyd and L. Vandenberghe, *Convex optimization*. Cambridge university press, 2004.
- [20] R. J. Glauber, “Time-dependent statistics of the ising model,” *Journal of mathematical physics*, vol. 4, no. 2, pp. 294–307, 1963.
- [21] J. R. Norris, *Markov chains*, vol. 2. Cambridge university press, 1998.
- [22] S. N. Ethier and T. G. Kurtz, *Markov processes: characterization and convergence*. John Wiley & Sons, 2009.
- [23] D. Bertsekas, *Dynamic programming and optimal control: Volume I*, vol. 4. Athena scientific, 2012.
- [24] M. Mezard and A. Montanari, *Information, physics, and computation*. Oxford University Press, 2009.
- [25] R. Van Handel, “Stochastic calculus, filtering, and stochastic control,” *Course notes.*, URL <http://www.princeton.edu/rvan/acm217/ACM217.pdf>, vol. 14, 2007.
- [26] N. G. Van Kampen, *Stochastic processes in physics and chemistry*, vol. 1. Elsevier, 1992.
- [27] P. C. Hohenberg and B. I. Halperin, “Theory of dynamic critical phenomena,” *Reviews of Modern Physics*, vol. 49, no. 3, p. 435, 1977.
- [28] J. Cardy, *Scaling and renormalization in statistical physics*, vol. 5. Cambridge university press, 1996.